Reg. No. | | | | | | | | | | |

[14CS43]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE–2016

MICROPROCESSORS AND MICROCONTROLLERS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full questions choosing at least TWO questions from each Part.
2. Any missing data may be suitably assumed

PART - A

1. a. What are the functions of below given Registers **08**

i. Ax

ii. Cx

iii. Dx

iv. BP

b. What are assembler directives? Explain the below given assembly directives. **06**

i. Extern

ii. Public

iii. Proc

2. a. Write an ALP to convert given BCD number into ASCII code. **06**

b. Briefly explain the following instructions: **08**

i. MOVSB

ii. CMP

iii. SH A

iv. LDS

3. a. What are different parameter passing techniques to a procedure? Explain with an example code. **08**

b. Write an ALP to check given password is correct or not by using string instructions. **06**

4. a. With neat pin diagram, explain the 8051 microcontroller. **08**

b. Explain the 8051 microcontroller RAM space allocation. **06**

PART - B

5. a. Assume that a 60Hz external clock is being fed into pin T-0 (P3.4). Write a 'C' program for counter 0 in Mode 2 to display the seconds and minutes on P1 and P2 respectively. **08**

b. Write the steps to program in Mode 2 timer mode. **06**

6. a. Briefly explain about SBUF and SCON registers in serial communication. **06**

b. Write an 8051 'C' program to send a message "Jain University" to a serial port. Assume XTAL=11.592 MHz, baud rate 9600, 8 bit data, 1 stop bit. **08**

7. a. Explain the Interrupts of 8051 microcontroller. **06**

b. Write a C program that continuously gets a single bit of data from P1.7 and send it to P1.0, while simultaneously creating a square wave of 200μs period on pin P2.3. Use timer 1 to create the square wave. Assume that XTAL = 11.0592 MHz. **08**

8. a. Write the steps required to program the ADC0808/0809. **06**

b. Write short notes on any two. **08**

i. Stepper motor

ii. PWM

iii. LCD